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**BS (Data Science) Part-III**
**DASC- 523 Parallel and Distributed Computing**

# LAB # 6 – Shared and Private variables in OpenMP

**Lab Objective: To understand the shared and private variables for threads in OpenMp.**

**Description:**

- In OpenMP, variables can be shared or private among threads.
- **Shared variables** are accessible by all threads. Any modification by one thread affects the others. They are declared using the shared clause.
- **Private variables** are unique to each thread. Each thread has its own copy, and changes are not visible to others. They are declared using the private clause.
- Default behavior:
- Global and static variables → shared
- Local variables inside parallel region → private

```
#pragma omp parallel shared (var1, var2) private (var3)
{
   // parallel region code
}
```

**Task – Demonstrating Shared variable**

```
#include <stdio.h>
#include <omp.h>

int main() {
omp_set_num_threads(5);
   int i;
   int x = 0;

   #pragma omp parallel shared(x)
   {
      int tid = omp_get_thread_num();
      x = x + tid;  // Shared variable modified by all threads
      printf("Thread %d updated x to %d\n", tid, x);
   }
```

```
    printf("\nFinal value of x (shared): %d\n", x);
    return 0;
}
```

**Task – Demonstrating Private variable**

```
#include <stdio.h>
#include <omp.h>

int main() {
omp_set_num_threads(5);
    int i;
    int x = 0;

    #pragma omp parallel private(x)
    {
        int tid = omp_get_thread_num();
        x = x + tid;     // Each thread has its own private copy
        printf("Thread %d has private x = %d\n", tid, x);
    }

    printf("\nFinal value of x (main thread): %d\n", x);
    return 0;
}
```